

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/4750/hw3-rha-sims-1`
Installed PrettyTables v3.0.11
Precompiling project...
 34576.8 ms   PrettyTables
 41667.9 ms   DataFrames
```

2795.6 ms Latexify → DataFramesExt
3 dependencies successfully precompiled in 81 seconds. 211 already precompiled.

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```

# Flows [m^3/d]
Q_river_inflow    = 100000.0
Q_waste_stream_1  = 10000.0
Q_waste_stream_2  = 15000.0

# Concentrations [mg/L]
DO_river_inflow   = 7.5
DO_waste_stream_1 = 5.0
DO_waste_stream_2 = 5.0

CBOD_river_inflow = 5.0
CBOD_waste_stream_1 = 50.0
CBOD_waste_stream_2 = 45.0

NBOD_river_inflow = 5.0
NBOD_waste_stream_1 = 35.0
NBOD_waste_stream_2 = 35.0

# River & Kinetic Properties
U = 6.0      # km/d
ka = 0.5     # d^-1 (reaeration)
kc = 0.3     # d^-1 (CBOD decay)
kn = 0.25    # d^-1 (NBOD decay)
Cs = 10.0    # mg/L (DO saturation)
dx2 = 15.0   # km between discharges

# --- Initial mixing with Waste 1 at x = 0 km
Q_segment_1 = Q_river_inflow + Q_waste_stream_1
CO_1 = (Q_river_inflow*DO_river_inflow + Q_waste_stream_1*DO_waste_stream_1) / Q_segment_1
BO_1 = (Q_river_inflow*CBOD_river_inflow + Q_waste_stream_1*CBOD_waste_stream_1) / Q_segment_1
NO_1 = (Q_river_inflow*NBOD_river_inflow + Q_waste_stream_1*NBOD_waste_stream_1) / Q_segment_1

# --- Analytic C(x) for a single segment that starts at x=0 with (CO,B0,NO)
function DO_analytic(x, CO, BO, NO, ka, kc, kn, Cs, U)
    alpha1 = exp(-ka * x / U)
    alpha2 = (kc/(ka - kc)) * (exp(-kc * x / U) - exp(-ka * x / U))
    alpha3 = (kn/(ka - kn)) * (exp(-kn * x / U) - exp(-ka * x / U))
    C = Cs*(1 - alpha1) + CO*alpha1 - BO*alpha2 - NO*alpha3
    return C
end

# --- Propagate segment 1 to x = 15 km (just upstream of discharge 2)

```

```

C_upstream_2 = DO_analytic(dx2, CO_1, BO_1, NO_1, ka, kc, kn, Cs, U)
B_upstream_2 = BO_1 * exp(-kc * dx2 / U)
N_upstream_2 = NO_1 * exp(-kn * dx2 / U)

# --- Mix with Waste 2 at x = 15 km (start of segment 2)
Q_segment_2 = Q_segment_1 + Q_waste_stream_2
CO_2 = (Q_segment_1*C_upstream_2 + Q_waste_stream_2*DO_waste_stream_2) / Q_segment_2
BO_2 = (Q_segment_1*B_upstream_2 + Q_waste_stream_2*CBOD_waste_stream_2) / Q_segment_2
NO_2 = (Q_segment_1*N_upstream_2 + Q_waste_stream_2*NBOD_waste_stream_2) / Q_segment_2

# --- Build C(x) from 0 to 50 km
x_initial = 0.0
x_final   = 50.0
points    = 1001
dx        = (x_final - x_initial) / (points - 1)

x_data = zeros(points)
C_data = zeros(points)

for i in 1:points
    x = x_initial + (i - 1) * dx    # <-- forward, not minus
    x_data[i] = x

    if x <= dx2
        # Segment 1: 0 ≤ x ≤ 15
        C_data[i] = DO_analytic(x, CO_1, BO_1, NO_1, ka, kc, kn, Cs, U)
    else
        # Segment 2: x > 15 (reset origin at 15)
        x_local = x - dx2
        C_data[i] = DO_analytic(x_local, CO_2, BO_2, NO_2, ka, kc, kn, Cs, U)
    end
end

# --- Find the minimum DO and where it occurs
minimum_C_1 = C_data[1]
initial_min = 1
for i in 2:points
    if C_data[i] < minimum_C_1
        minimum_C_1 = C_data[i]
        initial_min = i    # <-- update to the current index
    end
end
end

```

```

minimum_x_1 = x_data[initial_min]

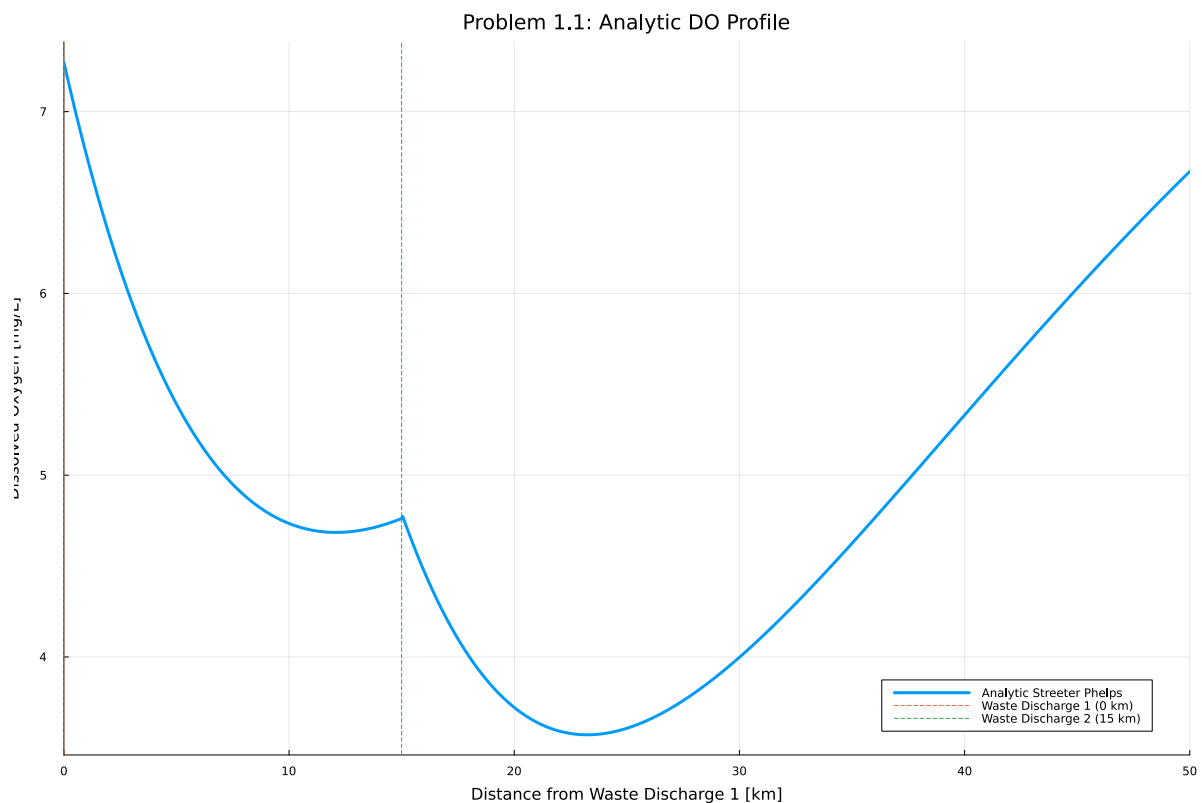
println("1.1 Minimum DO = ", round(minimum_C_1, digits=4), " mg/L at x = ", round(minimum_x_1, digits=2))

# --- Plot
p1 = plot(x_data, C_data;
          linewidth=3, label="Analytic Streeter Phelps",
          xlabel="Distance from Waste Discharge 1 [km]",
          ylabel="Dissolved Oxygen [mg/L]",
          title="Problem 1.1: Analytic DO Profile")

vline!([0.0]; linestyle=:dash, label="Waste Discharge 1 (0 km)")
vline!([dx2]; linestyle=:dash, label="Waste Discharge 2 (15 km)")
plot!(size=(1200, 800))
axis!((0, 50))
display(p1)

```

1.1 Minimum DO = 3.572 mg/L at x = 23.25 km



Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

```
function do_numerical(L, resolution, C0, B0, N0, ka, kn, kc, Cs, U)
    n = Int(L / resolution) # number of length steps
    C = zeros(n + 1)
    B = zeros(n + 1)
    N = zeros(n + 1)
    C[1] = C0
    B[1] = B0
    N[1] = N0
    for i = 1:n
        B[i+1] = B[i] * exp(-kc * resolution / U)
        N[i+1] = N[i] * exp(-kn * resolution / U)
        C[i+1] = C[i] + (resolution / U) * (ka * (Cs - C[i]) - kc * B[i] - kn * N[i])
    end
    return (C, B, N)
end

# Settings

resolution = 0.5 # km (delta X)
length_1 = dx2 # distance from 0 to 15 km
length_2 = 50.0 - dx2 # distance from 15 to 50 km

n1 = Int(round(L1 / resolution))
n2 = Int(round(L2 / resolution))
total_points = n1 + n2 + 1

# Initialize an array and fill in

x_num = zeros(total_points)
C_num = zeros(total_points)

# Segment 1 Integration

for i in 1:(n1+1)
    x_num[i] = (i-1) * resolution
```

```

    C_num[i] = C1[i]
end

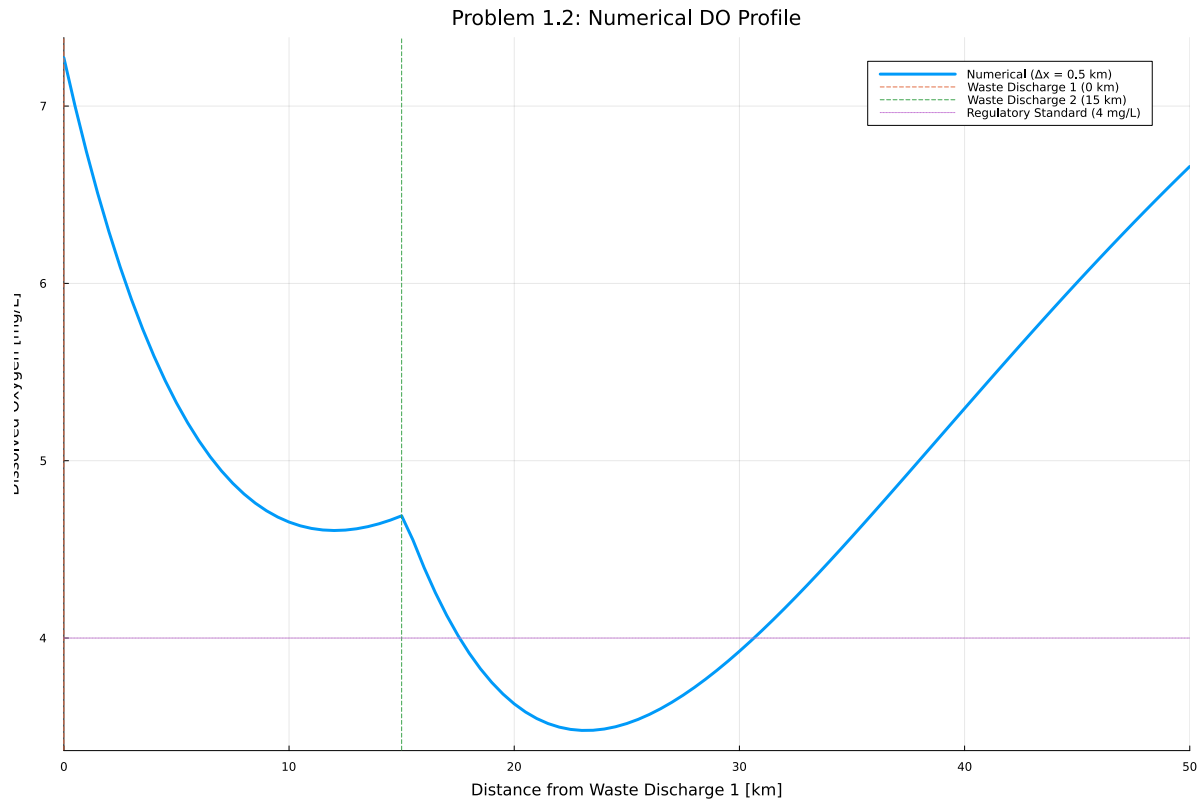
# Segment 2 Integration (after the 15 km point)

for i in 2:(n2+1)    # start at 2 to skip duplicate
    index = n1 + i
    x_num[index] = dx2 + (i-1) * resolution
    C_num[index] = C2[i]
end

# min
minimum_C_2, initial_min = findmin(C_num)
minimum_x_2 = x_num[initial_min]
println("1.2 Minimum DO = ", round(minC, digits=4), " mg/L at x = ", round(x_min, digits=2),

# plot
p2 = plot(x_num, C_num; lw=3, label="Numerical ( $\Delta x =$ $(resolution) km)",
    xlabel="Distance from Waste Discharge 1 [km]",
    ylabel="Dissolved Oxygen [mg/L]",
    title="Problem 1.2: Numerical DO Profile")
vline!([0.0]; linestyle=:dash, label="Waste Discharge 1 (0 km)")
vline!([dx2]; linestyle=:dash, label="Waste Discharge 2 (15 km)")
hline!([4.0]; linestyle=:dot, label="Regulatory Standard (4 mg/L)")
plot!(size=(1200, 800))
axis!((0, 50))
display(p2)

```



1.2 Minimum DO = 3.479 mg/L at $x = 23.0$ km

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

$$B = B_0 * \exp(-k_c * x / U)$$

$$N = N_0 * \exp(-k_n * x / U)$$

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss

whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

References

List any external references consulted, including classmates.